

Fig. 12: Complete impression of the circuit on the copper side

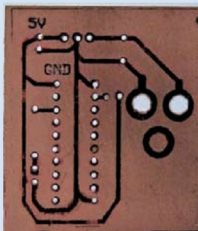


Fig. 14: IC holes drilled into the PCB



Fig. 13: 1mm hand PCB drill

Clearances for Electrical Conductors

Voltage (DC or Peak AC)	Internal	External (<3050m)	External (>3050m)
0-15V	0.05 mm	0.1 mm	0.1 mm
16-30V	0.05 mm	0.1 mm	0.1 mm
31-50V	0.1 mm	0.6 mm	0.6 mm
51-100V	0.1 mm	0.6 mm	1.5 mm
101-150V	0.2 mm	0.6 mm	3.2 mm
151-170V	0.2 mm	1.25 mm	3.2 mm
171-250V	0.2 mm	1.25 mm	6.4 mm
251-300V	0.2 mm	1.25 mm	12.5 mm
301-500V	0.25 mm	2.5 mm	12.5 mm

2. Rotate only the component by 90°

3. Move a selected item to the bottom copper layer

4. Move a selected item to the top copper layer

5. Component manager (opens the library window)

6. Options (can change the colour codes as well as all the dimensions and measurements)

7. Zoom to previous state

8. Zoom to fit the screen

9. Zoom out

10. Zoom in

11. Undo

12. Save file

13. Open file

14. General-purpose pointer, select and moving tool

15. Zoom into selected area

16. Place a pad (either through-hole or surface-mount pad). It is helpful in creating multilayer PCBs

17. Place a component. It will not open the library window

18. Place a trace or wire

19. Put a corner in a trace

20. Disconnect a trace

21. Place text in the layout

22. Place a rectangle

23. Place a filled power or ground plane

24. Place a circle or an arc

25. Insert an edge segment in the board perimeter

26. Pad information

27. Highlight net connections. For using this tool, you need to link the schematic of the circuit to this PCB. So go to File—>Link Schematic to PCB and choose the respective schematic file. Now while this tool is enabled, if you click a pin of the component it will show the pin to which it is connected. This is very helpful for complex circuits

28. Toggle display of the silkscreen layer

29. Toggle display of the top copper layer

30. Toggle display of the bottom copper layer

31. Toggle snapping to grid

PCB description. The circuit's PCB is described below, with the actual names of the components as in the library given within brackets:

1. At the centre is a 20-pin base (20 dip)

2. The 3-pin IC at the top is 7805 (semiconductor TO-220 without mounting hole, i.e., 7805 vreg)

3. On the left of 7805 is a 1000µF capacitor base (cap-lead spacing 4.5 mm (0.177 inch))

4. On the right of 7805 is a 0.1µF capacitor base (cap-lead spacing 2.5 mm (0.1 inch))

5. The three holes on the left are for the DC female input jack; in it 12V power supply will be connected

6. On the right of the 20-pin base is the base for a 1-kilo-ohm resistor (0.25-watt resistor with lead spacing of 0.35 inch)

7. Just below the 1-kilo-ohm resistor base is the LED base (LED-T1)

Rules of PCB making. 1. There should be no 90° connections of wires in the PCB. All connections should preferably be made at 45° (refer Fig. 3).

2. The mirror image of the PCB layout will always be imprinted on the copper side (the solder side of the copper-clad board) of the actual PCB. So for any text to be written on the copper side, the text should be mirrored during layout design so that the text